# Coupled Fluid-Elastic Interaction of a Hypersonic Airfoil

Undergraduate Thesis

*BITS 421T Thesis*

*by*

Vallabh Chandar

*Under the supervision of:*

Dr. Pardha Saradhi GV

And

Dr. Chennu Ranganayakulu




BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI, PILANI CAMPUS

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# Certificate

This is to certify that the thesis entitled, "Coupled Fluid-Elastic Interaction of a Hypersonic Airfoil" and submitted by VALLABH CHANDAR ID No. in partial fulfilment of the requirements of BITS F421T Thesis embodies the work done by him under my supervision.

_____

Supervisor: Dr Pardha Saradhi GV,

Assistant Professor,

Mechanical Engineering Department,

BITS Pilani, Hyderabad Campus

Date:

_____

Co - Supervisor: Dr Chennu Ranganayakulu,

Professor,

Mechanical Engineering Department,

BITS Pilani, Pilani Campus

Date:

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI,
PILANI CAMPUS

Abstract

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The coupled fluid-elastic interaction of hypersonic aerofoils is a complex and critical phenomenon that significantly influences the aerodynamic performance and structural integrity of hypersonic vehicles. This research aims to investigate and analyze the coupled fluid-elastic interaction to enhance our understanding of the underlying mechanisms and develop improved numerical models and analysis techniques.

The study employs a multidisciplinary approach, integrating principles from fluid dynamics, structural mechanics, and computational methods. Advanced computational fluid dynamics (CFD) techniques are utilized to model the highly unsteady and complex flow phenomena around hypersonic aerofoils.

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# Introduction

The coupled fluid-elastic interaction of hypersonic aerofoils is a challenging yet critical aspect of aerospace engineering, with profound implications for both aerodynamic performance and structural integrity. Hypersonic flight regimes, characterized by speeds exceeding five times the speed of sound, introduce unique fluid dynamic phenomena that significantly influence the behaviour of aerofoils. Simultaneously, the structural response of these aerofoils to aerodynamic forces become s increasingly important due to high temperatures, pressure loads, and dynamic loads. Understanding and effectively modelling the coupled fluid-elastic interaction is essential for the design, analysis, and optimization of hypersonic vehicles.

The primary objective of this thesis is to investigate and analyse the complex interactions between the fluid flow and structural response of hypersonic aerofoils. By comprehensively studying this coupled phenomenon, we aim to enhance our understanding of the underlying mechanisms, optimize aerodynamic performance, and ensure the structural integrity of these aerofoils in extreme operating conditions.

To accomplish this, a multidisciplinary approach will be adopted, integrating principles from fluid dynamics, structural mechanics, and computational methods. The thesis will begin with a comprehensive literature review, encompassing relevant research papers, technical reports, and theoretical frameworks related to hypersonic aerodynamics, aero elasticity, and fluid-structure interaction. This review will establish the foundation for the subsequent investigations.

Advanced computational fluid dynamics (CFD) techniques will be employed to accurately model the highly complex and unsteady flow phenomena around hypersonic aerofoils. This will enable the prediction of critical aerodynamic parameters, such as shock-wave location, boundary layer behaviour, and flow separation, which directly influence the aerodynamic performance of the aerofoil.

Simultaneously, Unsteady Reynolds-Navier Stokes equations will be utilized to analyse the structural response of the aerofoil to aerodynamic forces. The dynamic characteristics, such as natural frequencies, mode shapes, and stress distribution, will be investigated to understand the structural behaviour and identify potential points of vulnerability under extreme aerodynamic loading conditions.

The thesis will further focus on the development and implementation of advanced numerical algorithms to couple the fluid and structural solvers. This coupling will enable the simulation of the fluid-elastic interaction, allowing for a more accurate representation of the aerofoil's behaviour. The validation of these coupled numerical methods will be performed through comparison with experimental data, wind tunnel tests, and flight test data where available.

Additionally, the thesis aims to explore strategies for optimizing the aerodynamic performance and structural integrity of hypersonic aerofoils. This will involve the investigation of various design modifications, including shape optimization, material selection, and control surface manipulation. The goal is to enhance aerodynamic efficiency, reduce drag, improve stability, and ensure safe operation under extreme conditions.

In conclusion, this thesis seeks to advance our understanding of the coupled fluid-elastic interaction of hypersonic aerofoils by investigating its fundamental principles, developing accurate numerical models, and exploring optimization strategies. The findings from this research will contribute to the design and development of hypersonic vehicles, enabling improved aerodynamic performance, enhanced structural integrity, and safer operations in hypersonic flight regimes.

# Problem statement

Hypersonic flight poses unique challenges due to the complex interaction between the fluid flow and structural response of aerofoils operating at extreme speeds. Taking a missile as an example, travelling between Mach 2 and 3 it undergoes aerodynamic forces such as drag, etc. Now a missile will have to maintain a constant speed, even when it changes directions. But while changing directions the effect of aerodynamic may increase or decrease. This may have an impact on missile actuators which control the trajectory of the missile causing it to either deform or in the worst case break.

The coupled fluid-elastic interaction significantly impacts the aerodynamic performance and structural integrity of hypersonic vehicles. However, our understanding of this phenomenon is limited, and existing modelling approaches lack accuracy and robustness. Therefore, the problem statement of this research is to investigate and develop improved numerical models and analysis techniques for the coupled fluid-elastic interaction of hypersonic aerofoils, aiming to enhance aerodynamic performance and ensure structural integrity.

Key challenges that need to be addressed include:

- Unsteady and Complex Fluid Dynamics: Hypersonic flows exhibit shock waves, boundary layer transition, and flow separation, leading to non-linear and unsteady aerodynamic forces acting on the aerofoil. The accurate modelling and prediction of these complex flow phenomena, including shock wave location and intensity, are crucial for understanding the fluid-elastic interaction.

- Structural Response under Extreme Conditions: Hypersonic aerofoils experience high temperatures, pressure loads, and dynamic forces that can cause significant deformations, vibrations, and stress concentrations. The structural response must be accurately analysed to identify potential failure modes and ensure the aerofoil's structural integrity throughout its operational lifespan.

- Fluid-Structure Coupling: Existing numerical methods often lack a robust and accurate coupling between the fluid flow solver and the structural solver. The coupling of these two domains is essential to capture the mutual interactions and feedback mechanisms between the fluid forces and structural response accurately.

- Optimization of Aerodynamic Performance: Enhancing the aerodynamic performance of hypersonic aerofoils is crucial for achieving higher efficiency and stability. The problem of optimizing the aerofoil shape, control surface configuration, and other design parameters to minimize drag, maximize lift, and maintain stability in hypersonic flight regimes needs to be addressed.

- Validation and Experimental Correlation: The development and validation of numerical models for the coupled fluid-elastic interaction of hypersonic aerofoils require extensive experimental data and wind tunnel testing. The correlation between numerical simulations and experimental results is essential to validate and improve the accuracy of the models and analysis techniques.

Addressing these challenges will contribute to advancements in hypersonic vehicle design, leading to improved aerodynamic performance, enhanced structural integrity, and safer operations in extreme flight conditions. By developing accurate and robust numerical models for the coupled fluid-elastic interaction, engineers and researchers will be able to optimize aerofoil designs, improve flight efficiency, and ensure the structural integrity of hypersonic vehicles.

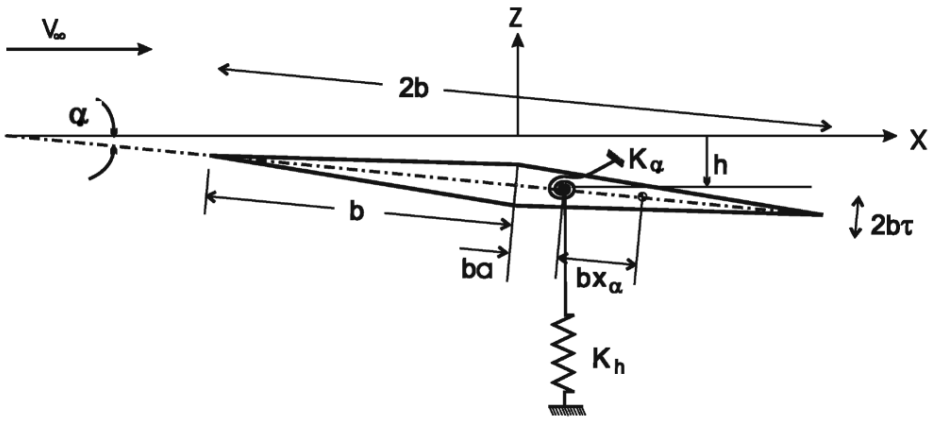



*Figure 1 A double wedged aerofoil with two springs connected, a linear and a torsional spring*

# Literature Review

The coupled fluid-elastic interaction of hypersonic aerofoils has been a subject of extensive research, aiming to understand and optimize the aerodynamic performance and structural integrity of hypersonic vehicles. The literature review provides an overview of key studies and advancements in this field, highlighting the fundamental principles, numerical modeling approaches, experimental investigations, and optimization strategies related to the coupled fluid-elastic interaction of hypersonic aerofoils.

### Fundamental Principles:

Anderson, J. D. (2006). Hypersonic and high-temperature gas dynamics. This foundational text provides comprehensive insights into hypersonic flow physics, including shock wave behaviour, boundary layer transition, and flow separation, which are crucial in understanding the fluid dynamics of hypersonic aerofoils.

### Numerical Modelling and Simulation Techniques:

- Jameson, A., Schmidt, W., & Turkel, E. (1981). "*Numerical solutions of the Euler equations by finite volume methods using Runge-Kutta time-stepping schemes*". This paper introduces the finite volume method and Runge-Kutta time-stepping schemes for solving the Euler equations, forming the basis for numerical simulations of hypersonic flows around aerofoils.
- Liu, C., Liu, Y., & Yang, Z. (2016). "*Fluid-structure interaction analysis of hypersonic flexible wings*". A study that investigates the fluid-structure interaction of hypersonic aerofoils using a coupled Euler solver and structural finite element analysis, highlighting the importance of accurate coupling methods for capturing the dynamic response.

Experimental Investigations:

- Howe, J. T. (2017). "*Experimental measurements in hypersonic aerodynamics*". This comprehensive review summarizes experimental techniques, such as wind tunnel testing and laser-based measurement methods, for obtaining aerodynamic data in hypersonic flow conditions, providing essential validation data for numerical models.
- Moore, T., & Carrico, J. (2012). "*A review of experimental hypersonic transition data and its application to flight vehicles*". This paper discusses experimental investigations on hypersonic boundary layer transition, offering valuable insights into the complex fluid behaviour that affects the structural response of aerofoils.

Aero elasticity and Structural Analysis:

- Hodges, D. H., & Pierce, G. A. (2011). "*Introduction to structural dynamics and aero elasticity*". This book provides a comprehensive introduction to aero elasticity and structural dynamics, covering key concepts such as flutter, divergence, and structural response analysis, essential for understanding the structural behaviour of hypersonic aerofoils.
- Saeed, F., & Gürdal, Z. (2018). "*Aero elastic analysis and optimization of hypersonic vehicles: A review*". A review paper that discusses various numerical and analytical methods for aero elastic analysis and optimization of hypersonic vehicles, emphasizing the importance of structural stability and control in the design process.

Optimization Strategies:

- Li, J., Li, Y., & Zhang, C. (2019). "*Aerodynamic shape optimization of hypersonic vehicles: A review"*. This review paper discusses different optimization methods, including gradient-based and evolutionary algorithms, applied to hypersonic vehicle shape optimization to improve aerodynamic performance and reduce drag.

- Rolt, A., & Hodges, D. H. (2014). "*Recent advances in aero elastic optimization with applications*". This paper presents recent advancements in aero elastic optimization techniques, including multidisciplinary optimization and uncertainty quantification, providing insights into the application of these methods to hypersonic aerofoil design.

The literature review showcases the significant progress made in understanding the coupled fluid-elastic interaction of hypersonic aerofoils. It highlights the importance of accurate numerical modelling, experimental validation, and optimization strategies to optimize aerodynamic performance and ensure structural integrity in hypersonic flight conditions. These studies provide a solid foundation for further research and the development of improved numerical models and analysis techniques for coupled fluid-elastic interaction in the context of hypersonic aerofoils.

# Numerical Methods

## Unsteady Reynolds-Averaged Navier-stokes

URANS (Unsteady Reynolds-Averaged Navier-Stokes) simulations are commonly used numerical methods for studying the coupled fluid-elastic interaction of hypersonic aerofoils. URANS approaches aim to capture the unsteady flow behaviour while providing a computationally efficient solution compared to more computationally expensive methods such as Large Eddy Simulation (LES) or Direct Numerical Simulation (DNS).

In the context of the coupled fluid-elastic interaction, URANS simulations are employed to model the unsteady aerodynamic forces acting on the aerofoil due to the hypersonic flow. These simulations solve the Reynolds-averaged Navier-Stokes equations, which provide a time-averaged description of the flow field. The governing equations are typically augmented with turbulence models to account for the turbulent behaviour in the flow.

$$\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_i} + \frac{1}{\rho}\frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\frac{\mu}{\rho}\frac{\partial U_i}{\partial x_j}\right) + \partial \frac{-\overline{u_i' u_j'}}{\partial x_j}$$

Where $t$ is time, $p$ is pressure, $\rho$ is the density, $\mu$ is viscosity, and $U$ denotes the velocity.

By incorporating the aero elastic response of the aerofoil, URANS simulations can assess the dynamic interaction between the fluid flow and structural deformation. The structural response is typically obtained through finite element analysis (FEA), considering the aero elastic properties of the aerofoil structure. The structural deformation is fed back to the URANS solver to compute the modified aerodynamic forces, creating a coupled fluid-structural simulation loop.

The URANS-based approach enables the investigation of aero elastic phenomena, such as flutter, divergence, and limit cycle oscillations, in hypersonic aerofoils. It allows for the assessment of the effects of fluid-structure interaction on the aerofoil's behaviour, including

changes in shock wave location, boundary layer behaviour, and dynamic pressure distribution. The obtained insights can inform the design process, optimize the aerofoil shape, and ensure safe operation under extreme hypersonic flight conditions.

It is important to note that URANS simulations have limitations. They are based on Reynolds-averaging assumptions, which neglect some of the unsteady flow phenomena and turbulence scales that may be significant in hypersonic flows. Additionally, accurately capturing the coupled fluid-elastic interaction requires robust and accurate coupling algorithms, and validation against experimental data and higher-fidelity simulations such as LES or DNS is crucial.

To overcome these limitations, researchers are exploring hybrid approaches that combine URANS simulations with more detailed turbulence models or high-fidelity methods for specific flow regions of interest. These hybrid methods aim to strike a balance between computational efficiency and accuracy in capturing the coupled fluid-elastic interaction of hypersonic aerofoils.

Overall, URANS simulations play a vital role in studying the coupled fluid-elastic interaction of hypersonic aerofoils, providing insights into the aerodynamic performance and structural behaviour of these complex systems. Further advancements in numerical techniques and validation against experimental data will continue to enhance our understanding and predictive capabilities in this field.

# 6DOF method

A 6DOF (Six Degrees of Freedom) solver and URANS (Unsteady Reynolds-Averaged Navier-Stokes) simulations are two complementary approaches used in the study of aerodynamics and flight dynamics of aerospace systems.

It is a mathematical model that describes the motion of an object in three translational and three rotational degrees of freedom. It considers forces and moments acting on the object, including aerodynamic forces, gravitational forces, and control inputs. The solver tracks the position, velocity, and orientation of the object over time and predicts its flight trajectory and attitude.

A 6DOF solver is commonly used in aerospace engineering for various applications, such as aircraft flight simulations, missile guidance, and spacecraft trajectory analysis. It provides a simplified representation of the system dynamics and allows for efficient computation of the vehicle's motion and response under different conditions.

On the other hand, URANS simulations are computational fluid dynamics (CFD) simulations that solve the Reynolds-averaged Navier-Stokes equations to model the flow behaviour around an object. URANS simulations are widely used to analyse unsteady flows, including the unsteady aerodynamic forces acting on aerospace vehicles.

URANS simulations provide valuable insights into the time-averaged flow properties, such as pressure distribution, velocity field, and flow separation, which are crucial in understanding the aerodynamic behaviour of the system. They are particularly useful in predicting the aerodynamic forces and moments acting on the object and capturing the unsteady flow phenomena that can impact the vehicle's stability and performance.

In aerospace applications, the 6DOF solver and URANS simulations are often used together to study the coupled behaviour of the vehicle's motion and the aerodynamic forces it experiences. The 6DOF solver provides the dynamic response of the vehicle based on the forces and moments obtained from the URANS simulations.

This coupling allows for a comprehensive analysis of the system's behaviour, considering the effects of unsteady aerodynamics on the vehicle's motion and stability. It enables the evaluation of the impact of changes in the flow field on the vehicle's trajectory, attitude, and overall flight performance.

By combining the 6DOF solver and URANS simulations, engineers can assess the coupled fluid-structural dynamics of aerospace systems and optimize their design for improved performance, stability, and control. This integrated approach facilitates the investigation of flight characteristics, manoeuvrability, and the influence of aerodynamic forces on the vehicle's response in real-world scenarios. Given below is the governing equation for the translational motion of the centre of gravity is solved for in the inertial coordinate system:

$$\dot{\vec{v}}_G = \frac{1}{m} \sum \vec{f}_G$$

Where $\dot{\vec{v}}_G$ is the translational velocity and $\vec{f}_G$ is the force vector.

The angular motion of the object $\dot{\vec{\omega}}_B$ which is computed through

$$\dot{\vec{\omega}}_B = L^{-1} \left( \sum \vec{M}_B - \vec{\omega}_B \times L\vec{\omega}_B \right)$$

Where $L$ is the inertia tensor, $\vec{M}_B$ is the momentum vector and $\vec{\omega}_B$ is the angular velocity vector of the rigid body.

Both angular and translational velocities are used in the dynamic mesh calculations to update the rigid body position

# Algorithm

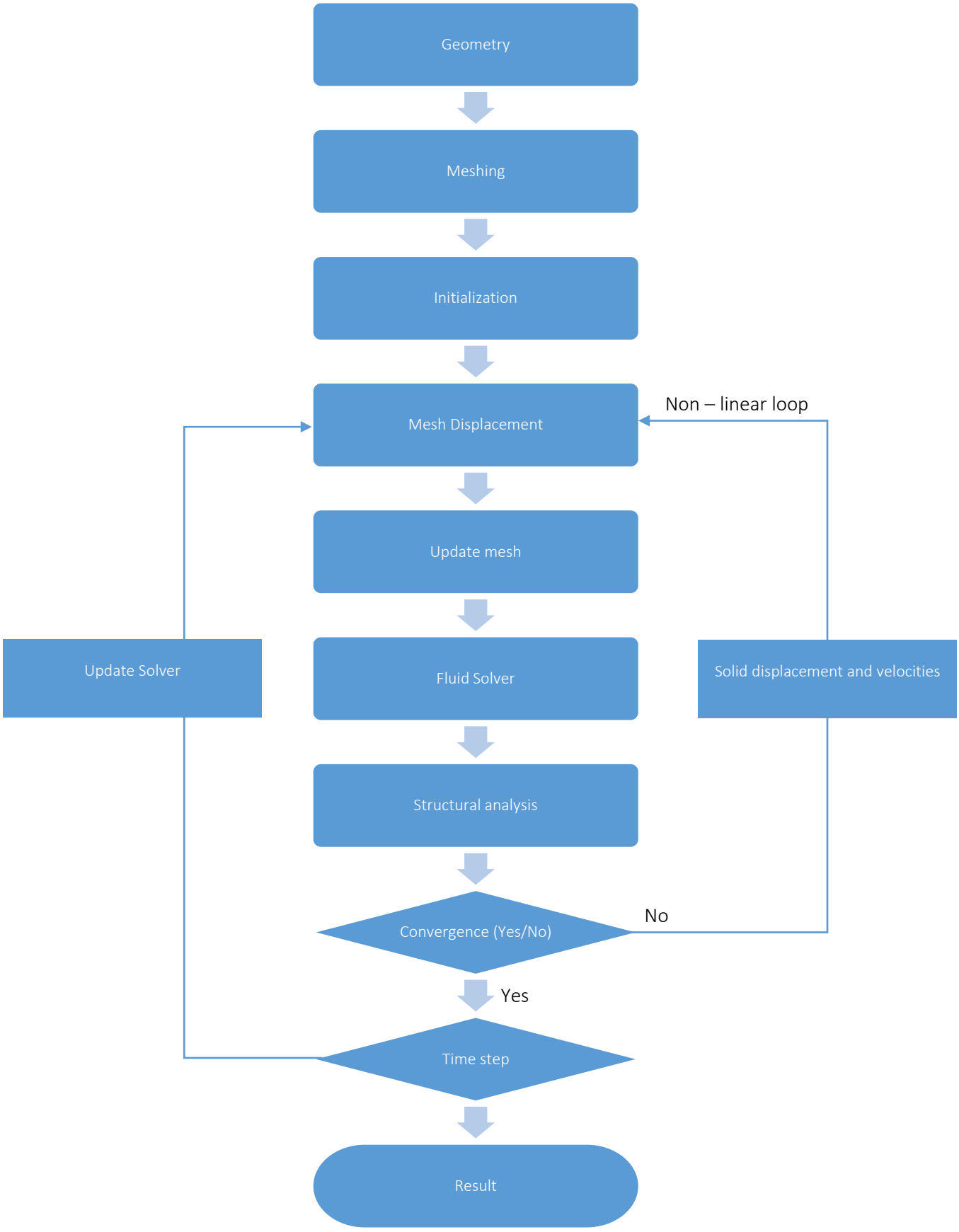



*Figure 9 Algorithm flow chart*

# Grid Expansion

Hypersonic flows are characterized by highly unsteady and three-dimensional phenomena, including shock waves, boundary layer transition, and flow separation. Additionally, the structural response of aerofoils to these aerodynamic forces can exhibit intricate deformations and oscillations.

To adequately capture these phenomena, grid expansion techniques are employed in the computational fluid dynamics (CFD) simulations of hypersonic flows. These techniques involve refining the computational mesh in regions of interest, such as shock wave locations, boundary layer regions, and areas with high-pressure gradients. By increasing the mesh resolution, the grid expansion ensures that the flow features are accurately captured, and the aerodynamic forces acting on the aerofoil are properly represented.

Moreover, grid expansion is also relevant for the structural analysis of hypersonic aerofoils. The aero elastic response of the aerofoil, including flutter, divergence, and limit cycle oscillations, necessitates capturing the structural deformations and stresses accurately. Refining the mesh in critical areas, such as near control surfaces or regions prone to high structural loads, enables a more precise representation of the structural behaviour.

By employing grid expansion techniques in both the fluid and structural domains, the coupled fluid-elastic interaction of hypersonic aerofoils can be more accurately captured and analysed.

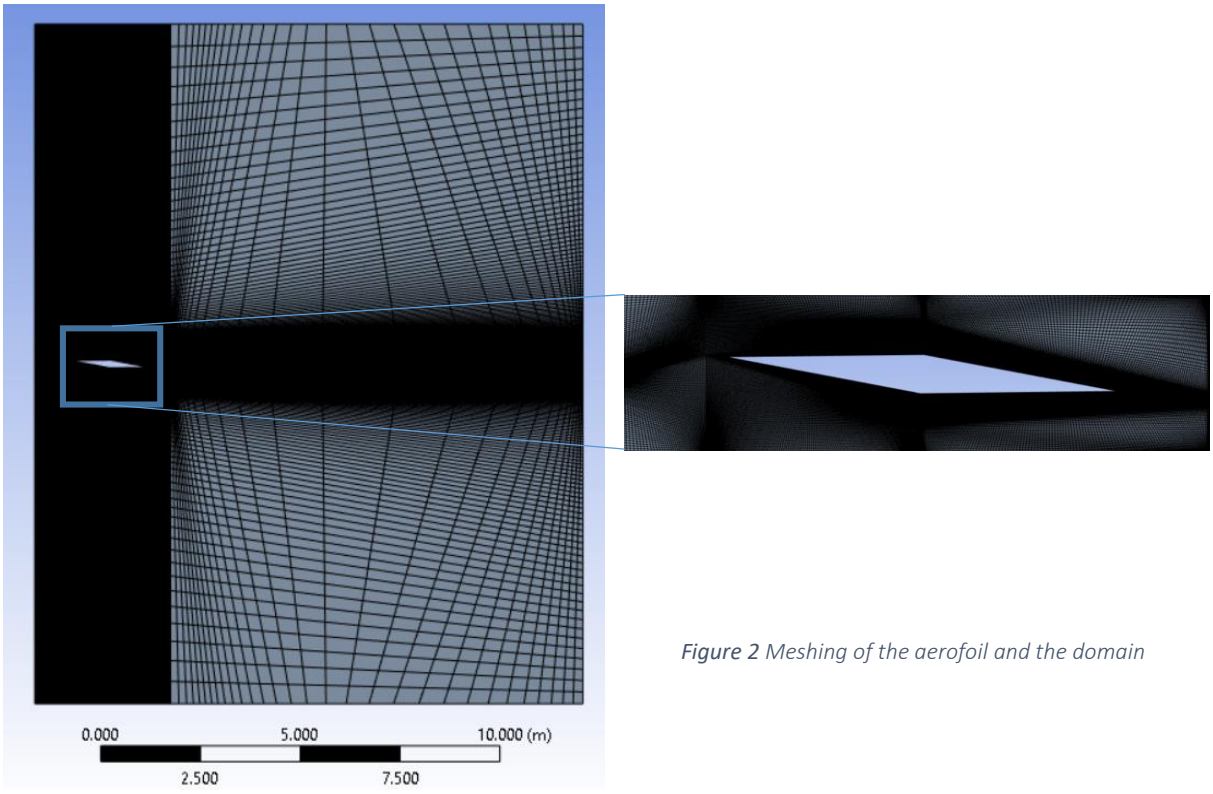



*Figure 2 Meshing of the aerofoil and the domain*

# User Defined Function (UDF)

A user-defined function (UDF) can be employed to enhance the numerical modelling and analysis. A UDF is a customized code written by the user to define specific boundary conditions, material properties, or physics-related equations within a computational fluid dynamics (CFD) simulation. For this particular case to model the torsion and other factors for simulation, a UDF is employed while using the 6DOF method during dynamic meshing. A sample code of the UDF is given below.

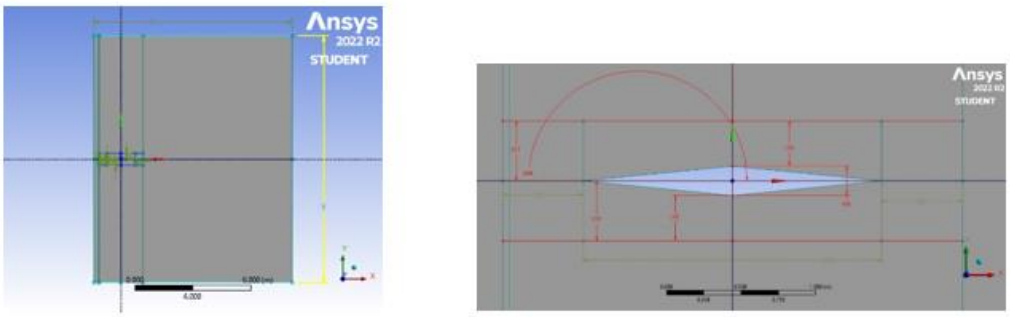
```
#include "udf.h"

static real velx = 0.0;
DEFINE_CG_MOTION(box, dt, vel, omega, time, dtime)
{
      NV_S(vel, =, 0.0);
      NV_S(omega, =, 0.0);

      velx = 0.05 * sin(3.1415927 * time / 10.);
      vel[1] = velx;
      Message("Time=%f, vel[1]=%f/n", time, vel[1])

}
```

# Mesh Boundary Dimensions and known parameters




(a) (b)

*Figure 3 (a) Domain (b) aerofoil*

a) Domain
   - X = 13.7m
   - Y = 17m

b) Aero foil
   - V56 = 0.2m
   - H55 = 2m

Boundary conditions selected:

- Pressure: 20265 Pa
- Mach no: 2.5
- Turbulence model: K-epsilon Realizable
- Temperature: 218K
- $U_x$ = 600 m/s
- Altitude: 12000m

# Results

Pressure Distribution Vs Angle of Attack

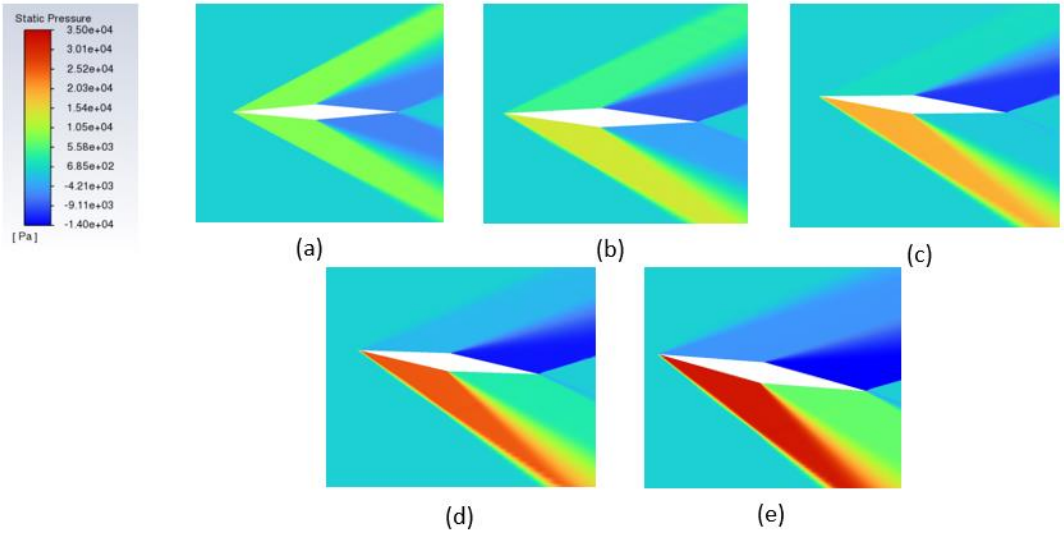



*Figure 4 Surface plot of pressure (a = 0°, b = 2.5°, c = 5°, d = 7.5°, e = 10°)*

Velocity Vs Angle of Attack

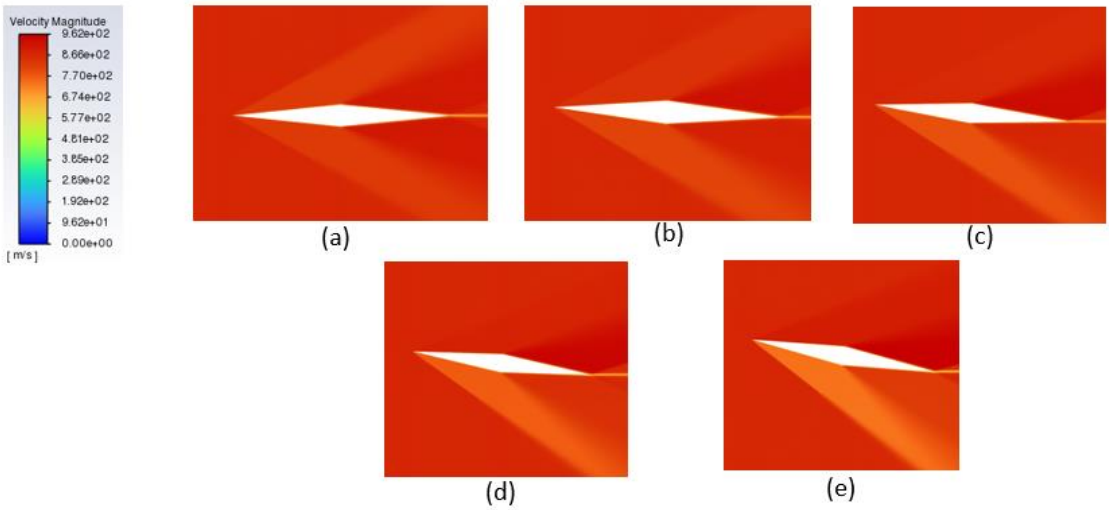



*Figure 5 Surface plot of velocity (a = 0°, b = 2.5°, c = 5°, d = 7.5°, e = 10°)*

Temperature Vs Angle of Attack

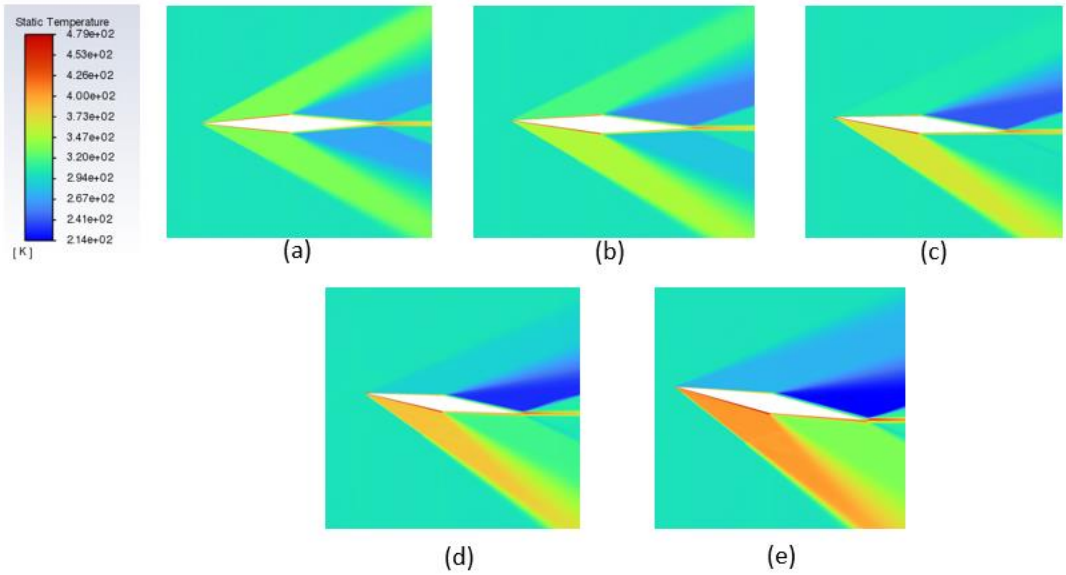



*Figure 6 Surface plot of temperature (a = 0°, b = 2.5°, c = 5°, d = 7.5°, e = 10°)*

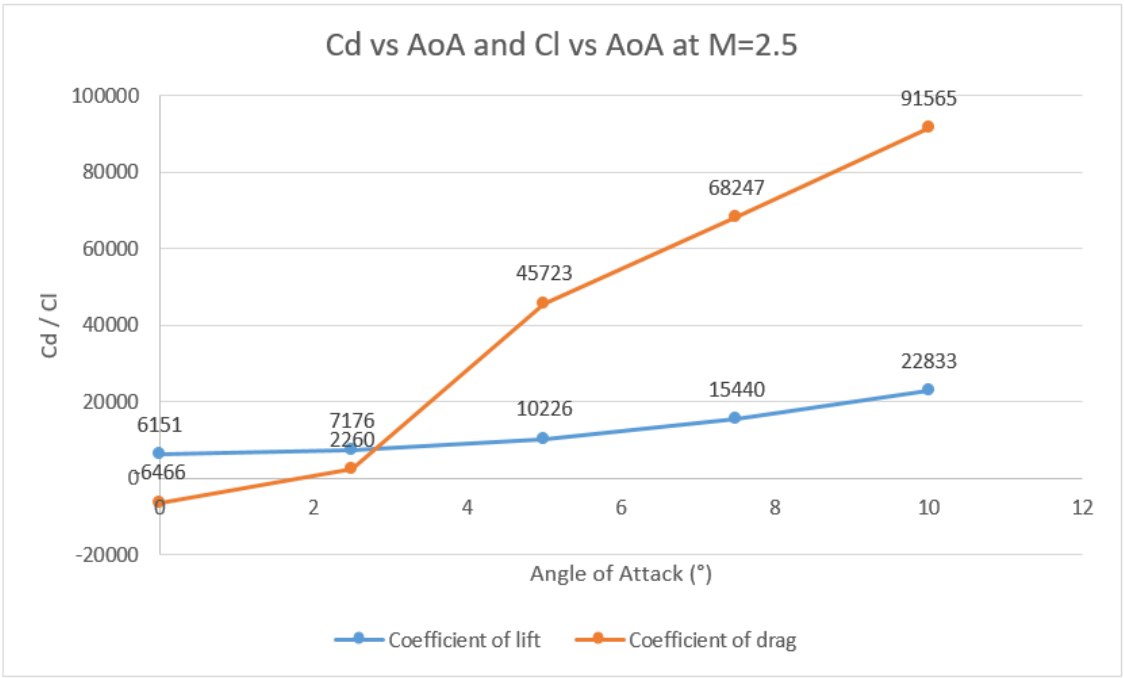



*Figure 7 Coefficient of Drag and Lift Vs AoA*

**Y+ values**

The Y+ value provides an indication of the mesh resolution in the boundary layer region, where viscous effects are significant. It is defined as the ratio of the distance of the first cell centroid to the wall (y) and the local viscous length scale (δ). It is commonly used to ensure that the mesh captures the near-wall flow physics accurately. The choice of appropriate Y+ values depends on the specific modelling approach and the turbulence model used in the CFD simulation. In this case we've tried to construct the mesh in such a way that the Y+ value remains between 15-250.

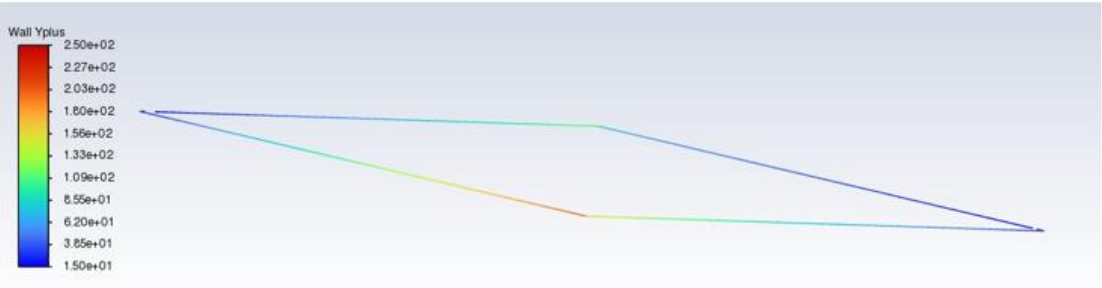



*Figure 8 Y+ plot around the aero-foil 15-250*

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